Light

GCSE Computer Science - Embedded Systems (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of an embedded system?

- ○ **A.** To run any software the user wants to install
- ○ **B.** To perform a specific dedicated function within a larger device
- ○ **C.** To replace all desktop computers
- ○ **D.** To provide internet access to buildings

Q2 (1 mark)

Which of these is NOT an embedded system?

- ○ **A.** A smartphone used for calls, apps, browsing and gaming

- B. A central heating thermostat
- C. A car parking sensor
- D. A lift controller

Q3 (1 mark)

A security alarm system in a house that detects intruders contains which type of system?

- A. A supercomputer
- B. A general-purpose server
- C. An embedded system dedicated to security monitoring
- D. No computer system

Q4 (1 mark)

Why can users not change the primary function of an embedded system?

- A. The hardware and software are designed specifically for one dedicated purpose
- B. Users are legally prohibited from touching embedded systems
- C. Embedded systems have no software at all
- D. All computers allow users to change their function

Q5 (1 mark)

Many embedded systems have no screen or display. Why is this acceptable?

- ○ **A.** Display technology has not been invented
- ○ **B.** All embedded systems must have large displays
- ○ **C.** Screens are illegal in embedded systems
- ○ **D.** Their automated functions often require no visual output to the user

Q6 (1 mark)

A car's cruise control system that maintains constant speed contains which type of system?

- ○ **A.** An embedded system controlling throttle based on speed sensors
- ○ **B.** A general-purpose laptop
- ○ **C.** No computer system
- ○ **D.** A gaming console

Q7 (1 mark)

Why do embedded systems typically need less memory than general-purpose computers?

- ○ **A.** Memory has not been invented for embedded systems
- ○ **B.** They run only specific software for their dedicated function, not multiple applications
- ○ **C.** All computers require exactly the same amount of memory

- ○ **D.** Embedded systems store more data than desktop computers

Q8 (1 mark)

Which capability do general-purpose computers have that embedded systems typically lack?

- ○ **A.** The ability to run multiple different user-installed programs
- ○ **B.** The ability to process data
- ○ **C.** The ability to use electricity
- ○ **D.** The ability to contain electronic circuits

Q9 (1 mark)

A games console controller (gamepad) contains which type of system?

- ○ **A.** A general-purpose workstation
- ○ **B.** A mainframe computer
- ○ **C.** An embedded system for detecting button presses and motion
- ○ **D.** No electronic components

Q10 (1 mark)

Embedded systems are often described as 'deterministic'. What does this mean?

- ○ **A.** They determine the weather forecast

- ○ **B.** Given the same inputs, they always produce the same outputs in the same time
- ○ **C.** They can only count numbers
- ○ **D.** They make random decisions each time they run

Q11 (1 mark)

Many embedded systems are designed to run continuously for years without restart. Why is this possible?

- ○ **A.** Their simple, dedicated software has fewer bugs and memory leaks than complex systems
- ○ **B.** Embedded systems never use any electricity
- ○ **C.** All computers can run forever without problems
- ○ **D.** Embedded systems do not contain any software

Q12 (1 mark)

A toaster with electronic browning control contains which type of system?

- ○ **A.** A general-purpose gaming PC
- ○ **B.** A server computer
- ○ **C.** An embedded system controlling heating elements and timing
- ○ **D.** A tablet computer

Q13 (2 marks)

Why must a car's anti-lock braking system (ABS) be a real-time embedded system?

- ○ **A.** Because it needs to display the current time to the driver
- ○ **B.** Because it requires an internet connection to function
- ○ **C.** Because it operates only during certain hours of the day
- ○ **D.** Because it must detect wheel lock-up and adjust brake pressure within milliseconds - any delay could cause loss of control

Q14 (2 marks)

Compare a dedicated e-book reader with a tablet computer in terms of embedded versus general-purpose characteristics.

- ○ **A.** Both are identical embedded systems
- ○ **B.** Both are identical general-purpose computers
- ○ **C.** The e-book reader is more embedded (dedicated to reading with fixed software) while the tablet is general-purpose (runs user-installed apps for various tasks)
- ○ **D.** Neither device contains any computer system

Q15 (2 marks)

Many embedded systems in industrial equipment operate reliably for 20+ years. Why can they achieve such long operational life?

- ○ **A.** Dedicated single-purpose design, thoroughly tested software, and no user modifications or software installations reduce failure points

- B. Industrial equipment is never actually used
- C. Long operational life is impossible for any computer system
- D. Industrial embedded systems are replaced every month

Q16 (2 marks)

An embedded system in a greenhouse controls ventilation fans and heaters. What type of components receive output from this embedded system?

- A. The system produces no outputs
- B. Only computer monitors and printers
- C. Only keyboards and mice
- D. Actuators such as motors (for fans) and heating elements - devices that create physical changes

Q17 (2 marks)

Embedded systems are found in billions of devices worldwide. What enables such massive production volumes?

- A. Embedded systems cannot actually be manufactured
- B. Each embedded system must be individually hand-crafted
- C. Standardised microcontroller chips, minimal component requirements, and simple dedicated designs enable low-cost mass production
- D. Production volumes are actually very low for embedded systems

Q18 (2 marks)

A railway level crossing that lowers barriers when trains approach contains which type of system?

- ○ **A.** A general-purpose laptop
- ○ **B.** An embedded system that detects trains and controls barrier motors with high reliability and safety
- ○ **C.** No computer system - only mechanical components
- ○ **D.** A smartphone app

Q19 (2 marks)

Why is debugging embedded systems often more challenging than debugging desktop software?

- ○ **A.** Limited debugging interfaces, real-time constraints, and hardware dependencies make it harder to observe and test behaviour compared to desktop environments with full debugging tools
- ○ **B.** Embedded systems have no bugs
- ○ **C.** Desktop software is always more complex than embedded systems
- ○ **D.** Debugging tools only exist for embedded systems

Q20 (2 marks)

Embedded systems often operate in harsh environments (extreme temperatures, vibration, dust). How do they achieve this?

- ○ **A.** Embedded systems cannot operate in harsh environments

○ **B.** General-purpose computers handle harsh environments better than embedded systems

○ **C.** Harsh environments are always avoided by locating embedded systems indoors

○ **D.** Rugged hardware design, no moving parts (no fans or hard drives), sealed enclosures, and industrial-grade components rated for extreme conditions

Q21 (2 marks)

An MRI scanner in a hospital contains embedded systems that control powerful magnets and process images. Why are embedded systems essential for this application?

○ **A.** General-purpose computers would work equally well for MRI control

○ **B.** MRI scanners do not require any computer systems

○ **C.** Real-time magnet control requires precise timing, high reliability is essential for patient safety, and dedicated systems ensure consistent operation

○ **D.** Embedded systems are used only because they are cheaper

Q22 (2 marks)

Many embedded systems can enter low-power 'sleep' modes and wake instantly when needed. Why is this capability important?

○ **A.** Sleep modes have no practical benefit

○ **B.** Battery-powered devices can extend operational life dramatically while remaining responsive, and mains-powered devices reduce energy costs and heat generation

- ○ **C.** Only general-purpose computers can enter sleep modes
- ○ **D.** Sleep modes make embedded systems unreliable

Q23 (2 marks)

Embedded systems typically store their software in non-volatile memory like flash or ROM. Why is this important?

- ○ **A.** The software remains intact when power is removed, allowing instant operation when powered on without reloading from external storage
- ○ **B.** Non-volatile memory is faster than all other types
- ○ **C.** Volatile memory would be more reliable
- ○ **D.** All computers store software identically

Q24 (3 marks)

A delivery drone contains multiple embedded systems for flight control, navigation, obstacle avoidance, and package handling. Analyse why separate embedded systems are used rather than one central computer.

- ○ **A.** Using separate systems is wasteful and provides no benefits
- ○ **B.** A single computer would be faster and more reliable
- ○ **C.** Drones are too simple to contain any computer systems
- ○ **D.** Separate systems provide fault tolerance (one failure does not crash the entire drone), allow specialised real-time processing for each function, enable parallel development by different teams, and isolate safety-critical flight control from less critical systems

Q25 (3 marks)

Smart electricity meters are embedded systems that have caused security concerns. Analyse the unique challenges of securing such widely-deployed connected embedded devices.

- ○ **A.** Smart meters face no security challenges because they are embedded
- ○ **B.** Challenges include millions of devices deployed for decades, limited processing for security features, physical accessibility enabling tampering, mandatory connectivity creating attack surfaces, and update difficulties across diverse installed hardware
- ○ **C.** General-purpose computers in homes would be more secure than smart meters
- ○ **D.** Security is irrelevant for utility meters

Q26 (3 marks)

Analyse why embedded systems are essential for mechanical ventilators that help patients breathe in intensive care.

- ○ **A.** General-purpose computers would be equally suitable for life-support ventilators
- ○ **B.** Cost is the only reason embedded systems are used
- ○ **C.** Ventilators do not require any computer control
- ○ **D.** Ventilators require extreme reliability (continuous operation without crashes), precise real-time control of breath timing and pressure, fail-safe designs with backup systems, and certified software that has been exhaustively tested for life-critical operation

Q27 (3 marks)

Modern precision agriculture uses embedded systems in tractors, drones, and sensors for automated farming. Analyse how these embedded systems work together.

- ○ **A.** Embedded systems cannot be used in agriculture
- ○ **B.** Soil sensors monitor moisture and nutrients (input embedded systems), tractors use GPS-guided embedded controllers for precise planting and fertilising (processing and actuation), drones with embedded vision systems survey crops, and farm management systems integrate data for optimised decisions - each specialised for its function
- ○ **C.** Only one type of embedded system is needed for all farming tasks
- ○ **D.** Traditional farming methods are impossible to improve with technology

Q28 (3 marks)

A company is designing a system for a remote oil pipeline monitoring station. Compare using embedded systems versus a general-purpose computer for this application.

- ○ **A.** Both options are identical for this application
- ○ **B.** Computer systems cannot operate in remote locations
- ○ **C.** General-purpose computers are always better for all monitoring applications
- ○ **D.** Embedded systems are superior for this application: they can operate in extreme temperatures, run for years without maintenance, consume minimal power (important for solar-powered remote sites), start instantly, and have no unnecessary components that could fail. General-purpose computers offer easier development but require climate control, maintenance, and more power

Q29 (3 marks)

Compare the testing methodology for an embedded system in a spacecraft versus a mobile phone app.

- ○ **A.** Both use identical testing approaches with no differences
- ○ **B.** Spacecraft embedded systems require radiation testing, extreme environment simulation, years of qualification, and mathematical proof of correctness since failures are unrecoverable. Apps use automated testing, beta releases, and quick patches since bugs cause inconvenience, not mission loss. The consequence of failure drives vastly different rigour levels
- ○ **C.** Mobile apps require more rigorous testing than spacecraft systems
- ○ **D.** Neither requires testing before release

Q30 (3 marks)

As embedded systems become more capable, they increasingly make autonomous decisions (self-driving cars, medical diagnosis assistants). Analyse the implications of this trend.

- ○ **A.** Autonomous embedded systems create no new challenges
- ○ **B.** Human oversight of technology is no longer needed
- ○ **C.** Embedded systems will never be capable of autonomous decisions
- ○ **D.** Increasing autonomy raises questions about accountability when systems make harmful decisions, requires new testing approaches for AI components, challenges traditional certification methods designed for deterministic systems, and demands consideration of ethical boundaries for machine decision-making while enabling beneficial applications impossible for humans alone

Total: **55** marks • Time: **30 min**

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